

Subrahmanya Rajesh Nayak

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EDUCATION

M.Sc. in Big Data and Artificial Intelligence

2024 - Present

SRH Berlin University of Applied Sciences

Grade: 1.5

B.E. in Information Science and Engineering

2020 - 2024

NMAM Institute of Technology, Karkala, India

CGPA: 8.37/10

Technical Skills

Programming Languages: Python, Java, R, SQL, C

Machine Learning & Data Science: TensorFlow, PyTorch, Scikit-learn, Keras, Pandas, NumPy, Hugging Face Transformers

Model Development & MLOps: TensorBoard, MLflow, Kubeflow, Airflow, Git, Docker, REST APIs

Big Data & Cloud Platforms: Apache Spark, Hadoop, Hive, Kafka, AWS, GCP, Azure

Databases: MySQL, MongoDB

Tools & Other Skills: Tableau, Power BI, Linux, Shell Scripting, Agile Methodologies, Data Visualization, NLP, Deep Learning

EXPERIENCE

Research Intern

Jan 2024 - June 2024

NITK Centre, Surathkal

Cybersecurity using Deep Learning(CAN Intrusion detection in automated vehicles)

- Worked on cybersecurity anomaly detection using **Python**, **TensorFlow**, and **Keras**.
- Applied **LSTM** models for intrusion detection on benchmark datasets.
- Improved model accuracy through preprocessing and hyperparameter tuning.
- Presented results biweekly to faculty supervisors.

Full Stack Developer Intern

Aug 2022 - Sep 2022

Zephyr Technologies, Remote

Online Quiz Platform

- Developed a full-stack quiz web app using **React**, **HTML**, **CSS**, **JavaScript**, and **Firebase**.
- Integrated dynamic quiz logic, timer, scoring, and user authentication.
- Worked with **Firebase Authentication** and **Realtime Database**.

Web and Mobile App Developer Intern

May 2022 - June 2022

RineX, Remote

Yoga Pose Guidance App

- Developed a yoga mobile app using **Android Studio** and **Java**.
- Implemented pose images, audio guidance, and timer features.
- Focused on intuitive UI/UX and smooth user navigation.

AI Intern

Feb 2021 - Mar 2021

Knowledge Solutions India (Microsoft Partner), Remote

AI-based Insurance Cost Prediction

- Built ML models using **Scikit-learn** and **XGBoost** for insurance claim predictions.
- Engineered features and handled missing data in large datasets.

- Created insightful visualizations using **Matplotlib** and **Seaborn**.

Projects

AQI Live Monitoring and Prediction

Python, Flask, Sklearn, HTML, CSS, JavaScript

- Developed a web-based dashboard to monitor real-time Air Quality Index using sensor data from API.
- Built a machine learning model to predict AQI levels using regression algorithms and environmental parameters.
- Deployed the system using Flask and integrated dynamic visualization for live AQI updates.

Deep Learning for Cataract Detection Using Fundus and Naked Eye Imaging

Python, TensorFlow, Keras, OpenCV, CNNs, DCNNs, Flask

- Designed and trained CNN-based models to detect cataract from both fundus and real-time naked eye images.
- Used data augmentation and preprocessing techniques to improve model accuracy and robustness.
- Achieved high performance on benchmark datasets and deployed a prototype system for real-world testing.

Object Detection and Control System

Python, OpenCV

- Developed a real-time object movement detection system using Python and OpenCV.
- Implemented automated scrolling triggered by directional object motion.
- Enabled conditional actions (e.g., alerts or system responses) based on specific movement patterns.
- Designed for minimal latency and responsive interaction with dynamic inputs.

PUBLICATIONS

- “Deep Learning Approaches for Cataract Detection Using Fundus and Real-Time Naked Eye Imaging”
Conference-2024 International Conference on Recent Advances in Science and Engineering Technology (ICRASET) [🔗 View Here](#)

LANGUAGES

English (Fluent), German (A2)

Hobbies

Problem solving, Traveling, Music production, Singing